

SAJJAD AHMAD

AI Engineer

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🔗 my-portfolio-mwyr.vercel.app

PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building AI applications using Python, Machine Learning, Computer Vision and Large Language Models. Experienced in developing RAG systems, AI agents, real-time face recognition solutions and deep learning projects using modern AI frameworks. Strong interest in solving real-world problems through practical AI development, with a continuous focus on learning, improving, and building reliable solutions.

EXPERIENCE

AI Engineer Intern

06/2025 – 09/2025

Alright Tech

- Developed AI and Generative AI applications using Python, LangChain, LangGraph, and CrewAI.
- Built Retrieval-Augmented Generation (RAG) applications by integrating Large Language Models (LLMs) with vector databases.
- Designed and implemented Multi-Agent AI systems to automate complex workflows and improve task execution.
- Worked with Hugging Face and PyTorch to develop and evaluate machine learning and deep learning models.
- Researched and implemented modern AI techniques, including prompt engineering, LLM integration, and AI workflow automation.

EDUCATION

Bachelor of Science in Computer Science

2022 – 2026

University of Peshawar

Peshawar, Pakistan

TECHNICAL SKILLS

Programming Languages

Python

Generative AI & LLMs

LangChain, LangGraph, Hugging Face Transformers, Prompt Engineering, RAG

DataBase

Mongodb, Vectordb, Memorysaver

AI/ML Frameworks

TensorFlow, PyTorch, Scikit-learn, Keras, OpenCV, streamlit, InsightFace, YOLO, FastAPI

Core Competencies

Machine Learning, Deep Learning, Neural Networks, NLP, Multi-Agent Systems, Data Analysis, Algorithm Design, Software Development, Problem Solving

PROJECTS

Multi-Agent University Assistant System

Stack: LangChain, LangGraph, Crewai, Nvidia-Nim

Developed an intelligent multi-agent AI system to automate academic support by resolving queries, managing schedules, and retrieving relevant resources. Leveraged LangChain, transformer-based NLP, and vector search to deliver faster, more accurate, and context-aware responses while improving the overall user experience.

AI Chatbot with Retrieval-Augmented Generation

Stack: LangChain, RAG, HuggingFace, Streamlit

Developed an intelligent conversational chatbot using Retrieval-Augmented Generation (RAG) to deliver accurate, context-aware responses from a domain-specific knowledge base. Integrated ChromaDB for semantic search and HuggingFace models with custom prompt engineering to improve retrieval quality and maintain context across multi-turn conversations.

Customer Churn Prediction

Stack: Scikit-learn, NumPy, Logistic Regression, Random Forest, Decision Trees

Developed a machine learning solution to predict customer churn using classification algorithms. Performed data preprocessing, feature engineering, model evaluation, and comparative analysis to identify the best-performing model while visualizing insights for data-driven decision-making.

Image Classification

Stack: PyTorch, OpenCV, CNN, ResNet

Developed a deep learning model for image classification using Convolutional Neural Networks (CNNs). Applied image preprocessing, data augmentation, and TensorFlow/Keras to train, evaluate, and optimize the model for accurate image recognition.

AI-Powered Face Attendance System

Stack: OpenCV, InsightFace, SQLite, InsightFace, Cosine Similarity

Developed a real-time face recognition and attendance management system using Python, OpenCV (cv2), InsightFace, FastAPI, SQLite, and React.js. Implemented accurate face detection, feature extraction, and recognition using InsightFace embeddings with cosine similarity matching. Designed a modular backend with REST APIs for student registration, attendance management, model training, and live recognition, while integrating a responsive React dashboard for monitoring and managing attendance records. The system supports automatic attendance marking, real-time webcam/IP camera streaming, embedding-based recognition, and scalable architecture for future deployment.

AI-Powered Real-Time Object Detection

Stack: PyTorch, OpenCV, Ultralytics YOLO, ByteTrack

Designed and implemented a real-time computer vision application using Ultralytics YOLOv8 and OpenCV. The system supports object detection from images, recorded videos, webcams, and IP cameras while displaying bounding boxes, class labels, confidence scores, FPS, and object counts. The project follows a clean, modular architecture for maintainability and future integration with APIs and web applications.